From Marchenko–Pastur to Woodbury: Direct Solvers for Long-Only Mean-Variance Portfolios

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Abstract

The sample covariance matrix of equity returns is severely ill-conditioned: a small number of pervasive risk factors dominate the spectrum while several hundred noise directions crowd the correlation bulk. Random matrix theory cleans the sample *correlation* $C = D^{-1/2}\hat{\Sigma}\, D^{-1/2}$: it identifies the signal factors via the Marchenko–Pastur bulk edge and replaces the noise eigenvalues with their average, yielding the trace-preserving "constant residual eigenvalue" (CRE) estimator $C_0 = \bar{\mu} I + U_k \Delta_k U_k^\top$, which on a five-year S&P 500 sample improves the condition number of the correlation by a factor of roughly 290 (to $\kappa \approx 313$).

The minimum-variance portfolio is solved in the standardised coordinates $y = D^{1/2}w$ (D the asset variances), an exact change of variables that poses the problem against C_0. We exploit its scalar-identity-plus-low-rank structure: the Woodbury identity gives the exact inverse of the active-set system at $O(n_a k^2 + k^3)$ per outer step of a primal-dual active-set method, replacing the $O(n_a^2 k + n_a^3)$ assemble-and-factor approach. The p linear balance constraints $Bw = c$ (the fully-invested budget $\mathbf{1}^\top w = 1$ as the leading case) are eliminated analytically through a $p \times p$ Schur complement. The Woodbury solve requires fewer flops when $k < n_a$, which holds across essentially all of the long-only frontier. The requisite eigenpairs are computed once by a randomised SVD applied directly to the return matrix at $O(nTk)$, without forming $\hat{\Sigma}$, so that for a sweep of S related solves the preprocessing amortises to $O(nTk/S)$ per solve.

On efficient-frontier sweeps the measured advantage over the assemble-and-factor baseline is fivefold warm and roughly fifteenfold cold, growing with n, and an exact critical-line sweep built on the same kernel traces the frontier for the cost of a coarse grid. The returned weights satisfy the KKT optimality conditions to solver accuracy, the Marchenko–Pastur threshold is shown to be an insensitive hyperparameter, and rolling out-of-sample backtests on two universes confirm that the cleaned estimator carries no measurable statistical penalty relative to standard shrinkage.

1 Introduction

The minimum-variance portfolio is the solution to a quadratic programme whose inputs are a covariance estimate and a set of weight constraints. In the large-n regime of institutional equity portfolios ($n/T \approx 0.4$ for a five-year daily window), the sample covariance $\hat{\Sigma} = X^\top X/T$ is nearly singular, with condition number $\kappa(\hat{\Sigma}) \approx 106{,}000$ on S&P 500 data. This ill-conditioning makes the portfolio optimisation problem numerically fragile.

Random matrix theory (RMT) attributes most of this ill-conditioning to noise in the *correlation* structure: applied to the sample correlation $C = D^{-1/2}\hat{\Sigma}\, D^{-1/2}$ ($D = \operatorname{diag}(\hat{\Sigma})$), the Marchenko–Pastur (MP) law identifies correlation eigenvalues below the bulk edge $\hat{\lambda}_+ = (1 + \sqrt{n/T})^2$ as noise and those above as genuine factor structure. The "constant residual eigenvalue" (CRE) cleaning method [8, 2, 1] replaces the noise eigenvalues with their average $\bar{\mu}$ (preserving $\mathrm{tr}(C) = n$) and produces the cleaned correlation

$$C_0 = \bar{\mu}\, I + U_k(\Lambda_k - \bar{\mu} I)U_k^\top,$$

where (U_k, Λ_k) are the k correlation eigenpairs above $\hat{\lambda}_+$. On S&P 500 data, $k = 17$ and $\kappa(C_0) \approx 313$, roughly 290 times better conditioned than the sample correlation ($\kappa(C) \approx 91{,}000$).

Implementations of CRE cleaning in the public domain typically form the cleaned *covariance* as a dense $n \times n$ matrix and pass it to a general-purpose quadratic-program solver. This paper exploits the scalar-identity-plus-low-rank structure of C_0, which renders the matrix inverse available in closed form via the Woodbury identity, without assembling either the $n \times n$ covariance or the $n_a \times n_a$ active-set restriction. Solving in the standardised coordinates $y = D^{1/2}w$ (exact, since $w \geq 0 \Leftrightarrow y \geq 0$) poses the minimum-variance problem against C_0 directly, so the solver inherits the cleaned correlation's conditioning. The result is a direct active-set portfolio solver substantially faster than the KKT-Cholesky approach of assembling the restricted system and factoring. Exploiting factor structure in portfolio quadratic programmes via the Sherman–Morrison–Woodbury identity goes back at least to Perold [18]; the contribution here is the combination with RMT-determined structure, randomised preprocessing, and an end-to-end complexity and accuracy account.

Contributions This paper makes three algorithmic contributions and one empirical contribution.

First, a direct Woodbury portfolio solver. The scalar-identity-plus-low-rank form of C_0 renders the active-set inner solve available in closed form via the Woodbury identity, at $O(n_a k^2 + k^3)$ per outer step in place of the $O(n_a^2 k + n_a^3)$ assemble-and-factor cost, without forming either the $n \times n$ covariance or its $n_a \times n_a$ active-set restriction (Section 4). The p linear balance constraints $Bw = c$ (the budget $\mathbf{1}^\top w = 1$ as the leading case) are eliminated analytically through a $p \times p$ Schur complement. We derive the exact crossover condition against KKT-Cholesky (Proposition 4.2): the Woodbury inner solve is cheaper per outer step precisely when $k < n_a$, a condition satisfied throughout the long-only equity regime (k/n_a between roughly 0.2 and 0.7 in our experiments), and we account for the total pipeline cost over a sweep of S related solves (Proposition 4.3), under which the $O(nTk)$ preprocessing amortises to $O(nTk/S)$ per solve.

Second, randomised-SVD preprocessing. The naive path forms $\hat{\Sigma}$ and takes a dense eigendecomposition, at $O(n^2T + n^3)$ cost and $O(n^2)$ storage. A randomised truncated SVD applied directly to X computes the top-k eigenpairs of $\hat{\Sigma}$ in $O(nTk)$ time and $O(nk)$ storage (Section 5); the measured portfolio impact of the approximation is below three basis points in any weight. For daily rolling windows we additionally implement an incremental rank-two update of the tracked eigenpairs that avoids re-sketching altogether.

Third, the scope of the structural trick (Remarks 2.2 and 4.4): ridge terms, diagonal loadings, externally specified factor models, and linear equality constraints preserve the Woodbury structure, whereas statistically richer cleanings (nonlinear shrinkage and general rotationally invariant estimators) do not.

Fourth, empirical validation. We compare Woodbury directly against KKT-Cholesky on the same RMT problem (Section 6), with a KKT optimality certificate for the returned weights and an exact critical-line sweep built on the same Woodbury kernel that traces every frontier breakpoint (Section 6.4). A k-sensitivity audit shows the MP threshold is not a sensitive hyperparameter (Section 6.5), and rolling out-of-sample backtests on two universes (S&P 500 and FTSE 100) show the cleaned estimator carries no measurable statistical penalty relative to standard shrinkage (Section 6.6).

Relation to existing work Laloux et al. [8] and Plerou et al. [19] brought the Marchenko–Pastur threshold to financial correlations; Bouchaud and Potters [1] and Bun, Bouchaud and Potters [2] develop the theory, including the statistically superior rotationally invariant estimators of Ledoit and Péché [11] and the nonlinear shrinkage of Ledoit and Wolf [12, 10]. Those modify the full spectrum and are not Woodbury-compatible; we work with CRE because clipping is the member of the family with exploitable structure (Remark 2.2), popularised for practitioners by López de Prado [13]. On the optimisation side, Perold [18] exploited diagonal-plus-low-rank factor structure in a portfolio QP, the critical line algorithm of Markowitz [15, 16, 20] traces the exact long-only frontier by active-set pivoting, and the randomised SVD is due to Halko, Martinsson and Tropp [5]. Our contribution is the pairing of RMT-determined low-rank structure with the active-set solve, the randomised preprocessing with an explicit pipeline complexity, and the measured end-to-end comparison.

This paper is self-contained: Algorithm 1 gives the complete primal-dual active-set solver including the Woodbury inner solve, and Section 5 develops the rSVD preprocessing independently.

Outline Section 2 constructs the cleaned estimator. Section 3 states the long-only problem. Section 4 develops the active-set reduction, the Woodbury portfolio algorithm, and the crossover analysis. Section 5 covers randomised SVD preprocessing. Section 6 reports timing, scaling, threshold sensitivity (Section 6.5), and out-of-sample results. Section 7 concludes.

2 The RMT-Cleaned Correlation Estimator

Let $X \in \mathbb{R}^{T\times n}$ be the demeaned return matrix for n assets over T periods, $\hat\Sigma = X^\top X/T$ the sample covariance, $D = \mathrm{diag}(\hat\Sigma)$ the diagonal of asset variances, and $C = D^{-1/2}\hat\Sigma\, D^{-1/2}$ the sample correlation. Individual variances are well estimated even in the high-dimensional regime; the noise that destabilises the portfolio lives in the *correlation* structure. Random-matrix cleaning is therefore applied to C, whose Marchenko–Pastur bulk is a clean band, and the cleaned covariance is rebuilt as $\Sigma_0 = D^{1/2}C_0D^{1/2}$. The minimum-variance problem is solved in the standardised coordinates $y = D^{1/2}w$ (Section 4), which recovers the scalar-identity Woodbury structure and its conditioning.

2.1 Marchenko–Pastur spectral separation

The Marchenko–Pastur (MP) law [14] describes the limiting spectral distribution of a sample correlation of standardised series when $n, T \to \infty$ with $n/T \to c \in (0, 1)$. Under the null of no correlation structure the bulk of the empirical spectral distribution is supported on $[\lambda_-, \lambda_+]$ where

$$\lambda_\pm = (1 \pm \sqrt{c})^2\,,$$

the unit-variance form ($\sigma^2 = 1$) that a correlation matrix enforces by construction, $\mathrm{tr}(C) = n$. In finite samples c is replaced by n/T, giving the empirical upper edge $\hat\lambda_+ = (1 + \sqrt{n/T})^2$.

Correlation eigenvalues above $\hat\lambda_+$ carry genuine factor structure; those below are consistent with noise and carry no reliable directional information. We denote by k the number of signal eigenvalues, $k = |\{j : \lambda_j(C) > \hat\lambda_+\}|$. For S&P 500 data with $n = 494$ and $T = 1213$, $\hat\lambda_+ \approx 2.28$ and $k = 17$; for the factor-calibrated synthetic model of Section 6 the detected k grows roughly as $n/24$ in this regime. Working with the correlation rather than the covariance matters here: the covariance mixes the (well-estimated) variances into the spectrum, so its bulk is not a clean MP band and the edge $\hat\lambda_+$ has no scale-free form.

2.2 Construction of C_0 and the cleaned covariance

Let $C = U\Lambda U^\top$ be the eigendecomposition of the sample correlation. Partition the eigenpairs by the MP edge: $U_k \in \mathbb{R}^{n\times k}$ holds the k signal eigenvectors with $\Lambda_k = \mathrm{diag}(\lambda_1, \ldots, \lambda_k)$, and U_0 the noise eigenvectors. The bulk eigenvalues are replaced by their average,

$$\bar\mu \;=\; \frac{1}{n-k}\sum_{j>k}\lambda_j \;=\; \frac{n-\sum_{j\le k}\lambda_j}{n-k}\,,$$

which preserves the trace $\mathrm{tr}(C) = n$. The resolution of the identity $U_kU_k^\top + U_0U_0^\top = I$ yields the "constant residual eigenvalue" (CRE), or eigenvalue-clipping, cleaned correlation of Laloux et al. [8] and Bun, Bouchaud and Potters [2] (popularised for practitioners by López de Prado [13]),

$$C_0 = U_k\Lambda_kU_k^\top + \bar\mu\, U_0U_0^\top = \bar\mu\, I + U_k\Delta_kU_k^\top\,, \qquad \Delta_k = \Lambda_k - \bar\mu I = \mathrm{diag}(\delta_1, \ldots, \delta_k)\,,$$

scalar-identity-plus-low-rank, with $\delta_j = \lambda_j - \bar\mu > 0$.

The cleaned covariance is $\Sigma_0 = D^{1/2}C_0D^{1/2}$. Because $D^{1/2}$ is positive-diagonal, $\Sigma_0 \succ 0$, and the change of variables $y = D^{1/2}w$ turns the minimum-variance problem $\min_w w^\top\Sigma_0 w$ subject to $Bw = c$, $w \ge 0$ into

$$\min_y\; y^\top C_0\, y \quad \text{subject to} \quad BD^{-1/2}\, y = c, \quad y \ge 0\,,$$

exactly (Section 4), since $w^\top\Sigma_0 w = y^\top C_0 y$ and $w \ge 0 \Leftrightarrow y \ge 0$. The solve therefore runs against the scalar-identity-plus-low-rank C_0, not the diagonal-plus-low-rank Σ_0.

Proposition 2.1 (Spectral properties of C_0)**.** *The cleaned correlation satisfies:*

1. *$C_0 \succ 0$ with eigenvalues $\{\lambda_1, \ldots, \lambda_k, \bar\mu, \ldots, \bar\mu\}$ and $\mathrm{tr}(C_0) = n$.*
2. *$\kappa(C_0) = \lambda_{\max}(C)/\bar\mu$.*
3. *$C_0 = C - U_0(\Lambda_0 - \bar\mu I)U_0^\top$: it differs from C only in the noise subspace.*
4. *The matrix-vector product $C_0v = \bar\mu v + U_k(\Delta_k(U_k^\top v))$ costs $O(nk)$ and requires $O(nk)$ storage.*

Proof. (1) follows from the eigendecomposition $C_0 = [U_k\ U_0]\,\mathrm{diag}(\Lambda_k, \bar\mu I_{n-k})\,[U_k\ U_0]^\top$, $\delta_j > 0$ for the signal eigenpairs, and the definition of $\bar\mu$; the trace is preserved. (2) the extreme eigenvalues are λ_1 and $\bar\mu$. (3) is direct algebra. (4) two dense products with $U_k \in \mathbb{R}^{n\times k}$. □

Remark 2.1 (Condition number)**.** *Cleaning lifts the noise floor of the correlation to $\bar\mu$ while preserving the signal, so $\kappa(C_0) = \lambda_{\max}(C)/\bar\mu \approx 313$ on S&P 500 data, against $\kappa(C) \approx 91{,}000$ for the sample correlation (and $\kappa(\hat\Sigma) \approx 106{,}000$ for the sample covariance): an improvement of roughly $290\times$. Solving in the standardised coordinates $y = D^{1/2}w$ inherits this conditioning. The rebuilt covariance Σ_0 is itself far worse conditioned ($\kappa(\Sigma_0) \approx 1{,}400$, inflated by the spread of asset variances, a factor of ≈ 49 between the largest and smallest on this sample), which is precisely why the solve is posed in y. The single noise floor also keeps the $k \times k$ Woodbury correction matrix well-conditioned regardless of n.*

Remark 2.2 (Relation to optimal RMT cleaning)**.** *CRE clipping is the simplest rotationally invariant estimator (RIE) and is not loss-optimal: nonlinear shrinkage [12, 10] and the oracle RIE of Ledoit–Péché and Bun et al. [11, 2], which reshape* every *eigenvalue, dominate it in Frobenius loss. We nevertheless use CRE for a structural reason: clipping is the only member of the family whose cleaned correlation is* scalar-identity-plus-low-rank*, the spiked form [7] that admits the closed-form*

Woodbury solve of Section 4; full-rank RIE modifications would require a general factorisation. Whether the statistical gap matters for long-only minimum variance is an empirical question, and on our data it is small (Section 6.6).

Remark 2.3 (Externally specified factor models)**.** *The Woodbury solver applies for any choice of k: a practitioner with an external factor model (a commercial risk model, or a PCA with k fixed by cross-validation) can supply k_{ext} eigenpairs directly, bypassing the MP threshold. What the solve requires is not the source of k but the replacement of the remaining $n-k$ correlation eigenvalues by a single scalar $\bar{\mu}$.*

2.3 Choice of k

The threshold k is determined by the MP upper edge $\hat{\lambda}_+$; on the S&P 500 sample it is stable near the central value $k = 17$. An audit at $k-1$, k, and $k+1$ adds negligible cost (one extra Woodbury solve per audit point) and reveals the portfolio's sensitivity to the threshold, which is small in the bulk of the frontier (Section 6.5).

Two biases are worth noting: finite-T Tracy–Widom fluctuations of the largest noise eigenvalue at scale $O(T^{-2/3})$ [7, 3], and diffuse sector structure just below the edge. Both perturb k by at most an eigenvalue or two, exactly what the $k \pm 1$ audit of Section 6.5 covers.

3 Problem Setup

The long-only mean-variance portfolio solves

$$\min_{w}\; w^\top \Sigma_0\, w - \rho\, \hat{\mu}^\top w \quad \text{subject to} \quad Bw = c, \quad w \geq 0\,,$$

where $\rho \geq 0$ is a return-tilt weight, $\hat{\mu}$ is a vector of expected returns, $B \in \mathbb{R}^{p\times n}$ of full row rank and $c \in \mathbb{R}^p$ carry $p \ll n$ linear balance constraints, and $\Sigma_0 = D^{1/2} C_0 D^{1/2}$ is the RMT-cleaned covariance rebuilt from the cleaned correlation C_0 of the previous section (Section 2). The fully-invested budget constraint $\mathbf{1}^\top w = 1$ is the case $p = 1$, $B = \mathbf{1}^\top$, $c = 1$; it is the leading case throughout, and the reader may substitute it wherever a balance system is not written out. General B additionally accommodates group budgets (sector sleeves summing to prescribed weights) and factor neutralities $\beta^\top w = 0$ at the same cost structure. The problem is solved in the standardised coordinates $y = D^{1/2} w$ against C_0 (Section 4); the map is exact and preserves the well-conditioned scalar-identity Woodbury structure.

Remark 3.1 (Why not blend with the sample correlation?)**.** *A shrinkage blend $(1-\alpha)C + \alpha C_0$ with $\alpha \in (0,1)$ would re-introduce a fraction of the bulk correlation eigenvalues that the cleaning replaces. Our reason for fixing $\alpha = 1$ is algebraic, not statistical: only at $\alpha = 1$ does the system matrix retain the scalar-identity-plus-low-rank form of C_0 that the Woodbury solve of Section 4 requires. For $\alpha \in (0,1)$ the restricted matrix $(1-\alpha)C_{\mathcal{A}} + \alpha C_{0,\mathcal{A}}$ has full-rank residual structure (no $k \times k$ correction matrix exists) and the $O(n_a k^2)$ cost advantage is lost. The statistical case for CRE cleaning rests on the RMT literature (Section 2); Section 6.6 reports out-of-sample evidence that it is competitive with standard shrinkage on the data used here.*

In the standardised coordinates the balance constraints $BD^{-1/2}y = c$ are eliminated analytically through a $p \times p$ Schur complement and the non-negativity constraint $y \geq 0$ is enforced by a primal-dual active-set outer loop, each step of which re-solves the balance-constrained subproblem on the current active set. Section 4 develops this reduction together with the Woodbury inner solve that is this paper's contribution.

4 The Woodbury Portfolio Solver

4.1 Primal-dual active-set reduction

By the change of variables $y = D^{1/2}w$ of Section 2 the minimum-variance problem is solved against the cleaned correlation C_0: minimise $y^\top C_0\, y - \rho\, \tilde{\mu}^\top y$ subject to $\tilde{B}y = c$, $y \geq 0$, with $\tilde{B} = BD^{-1/2}$ and $\tilde{\mu} = D^{-1/2}\hat{\mu}$, and the portfolio recovered as $w = D^{-1/2}y$. To lighten notation the rest of this section writes w, B, $\hat{\mu}$ for the standardised weight y, the transformed balance matrix $\tilde{B}$, and $\tilde{\mu}$; every displayed formula is in these standardised coordinates and the reported weights are $D^{-1/2}$ times the solve's output.

The balance constraints $Bw = c$ and non-negativity $w \geq 0$ are handled separately. Fix a set $\mathcal{A} \subseteq \{1, \ldots, n\}$ of active assets and write $B_{\mathcal{A}}$ for the columns of B indexed by $\mathcal{A}$. The balance-constrained subproblem on $\mathcal{A}$,

$$\min_{w_{\mathcal{A}}:\, B_{\mathcal{A}} w_{\mathcal{A}} = c} \; w_{\mathcal{A}}^\top C_{0,\mathcal{A}}\, w_{\mathcal{A}} - \rho\, \hat{\mu}_{\mathcal{A}}^\top w_{\mathcal{A}}\,,$$

has, by the KKT conditions, the closed-form solution

$$w_{\mathcal{A}}^\star \;=\; \tfrac{1}{2} V_1\, \lambda + \tfrac{\rho}{2} v_2\,, \qquad V_1 = (C_{0,\mathcal{A}})^{-1} B_{\mathcal{A}}^\top \in \mathbb{R}^{n_a \times p}\,, \quad v_2 = (C_{0,\mathcal{A}})^{-1}\hat{\mu}_{\mathcal{A}}\,,$$

where the p balance multipliers $\lambda \in \mathbb{R}^p$ are recovered analytically from feasibility $B_{\mathcal{A}} w_{\mathcal{A}}^\star = c$ through the $p \times p$ Schur complement

$$S\,\lambda = 2c - \rho\, B_{\mathcal{A}} v_2\,, \qquad S = B_{\mathcal{A}} V_1 = B_{\mathcal{A}} (C_{0,\mathcal{A}})^{-1} B_{\mathcal{A}}^\top \succ 0\,,$$

which is SPD because $C_{0,\mathcal{A}} \succ 0$ and $B_{\mathcal{A}}$ has full row rank; the dense $p \times p$ solve is negligible for the small p typical of balance conditions. In the budget case ($p = 1$, $B_{\mathcal{A}} = \mathbf{1}^\top$, $c = 1$) this collapses to a single vector $v_1 = (C_{0,\mathcal{A}})^{-1}\mathbf{1}$, a scalar Schur complement $S = \mathbf{1}^\top v_1$, and $w_{\mathcal{A}}^\star = \theta v_1 + \frac{\rho}{2} v_2$ with $\theta = \frac{\lambda}{2} = (1 - \frac{\rho}{2}\mathbf{1}^\top v_2)/(\mathbf{1}^\top v_1)$.

Non-negativity is enforced by the primal-dual active-set outer loop of the companion paper [21]. Throughout, the objective gradient is

$$g \;=\; 2\, C_0 w - \rho\, \hat{\mu}\,,$$

evaluated through the low-rank form $C_0 w = \bar{\mu} w + U_k(\Delta_k(U_k^\top w))$ at cost $O(nk)$ (Proposition 2.1); the return matrix X never enters the solve. Stationarity of the restricted subproblem reads $g_j = (B_{\mathcal{A}}^\top \lambda)_j$ for every $j \in \mathcal{A}$, with the multipliers λ delivered for free by the Schur solve above. In the budget case $B = \mathbf{1}^\top$ all active gradients coincide with the single multiplier $\lambda = 2\theta$, which the implementation then reads off as the median of g_j over $\mathcal{A}$ (used for robustness, exact up to roundoff).

The loop drops any active asset whose weight turns negative (the *primal step*) and re-admits any excluded asset whose reduced gradient $g_i - (B^\top\lambda)_i$ is negative (the *dual step*), re-solving the restricted subproblem after each exchange and stopping once $w_{\mathcal{A},i} \geq -\varepsilon$ for all $i \in \mathcal{A}$ and $g_i \geq (B^\top\lambda)_i - \varepsilon$ for all $i \notin \mathcal{A}$. In the budget case $(B^\top\lambda)_i = \lambda$ reduces to the scalar threshold $\nu = 2\theta$. These toggles are the principal pivots of the P-matrix linear complementarity problem $(C_0, -\rho\hat{\mu})$: because $C_{0,\mathcal{A}} = \bar{\mu} I_{n_a} + U_{k,\mathcal{A}} \Delta_k U_{k,\mathcal{A}}^\top \succ 0$ for every $\mathcal{A}$ (all $\delta_j > 0$ and $\bar{\mu} > 0$), every principal submatrix is invertible, each pivot is well defined, and the restricted subproblem has a unique optimum.

The outer loop, its finite-termination proof, and the batch/least-index pivot schedule that makes it robust are developed in full generality in the companion paper [21], of which the RMT problem is the instance $A = C_0$, $b = \rho\hat{\mu}$. Theorem 4.1 there gives finite termination at the unique global minimiser with no non-degeneracy assumption, and licenses any *direct* factorisation of the free-set system as the inner solve: the Woodbury solve developed below is exactly such a solver, so termination is inherited.

Remark 4.1 (Implementation safeguards)**.** *The reference implementation runs the batch fast path (assets whose weights are strongly negative, $w_i < -10\,\varepsilon$, are dropped together before any single-exchange step) and carries an iteration cap and an active-set-change check as a lightweight stand-in for the least-index fallback of [21]. In all experiments of Section 6 neither safeguard triggered and the batch drops did not affect the returned portfolio (the converged weights agree with the KKT-Cholesky reference to machine precision, Section 6.1).*

Remark 4.2 (Tolerance scale)**.** *The tolerance ε applies to the standardised weights (primal step) and the reduced gradient (dual step). In the standardised coordinates both live on the $O(1)$ correlation scale, with noise floor $\bar{\mu} \approx 0.5$, so the default $\varepsilon = 10^{-6}$ is a standard KKT tolerance; the change of variables removes the dependence on the raw return scale that a covariance formulation would carry.*

What remains is the inner solve $(C_{0,\mathcal{A}})^{-1}b$ required at each outer step. This inverse is available in closed form via the Woodbury identity, reducing the dominant per-step cost from $O(n_a^2 k + n_a^3)$ to $O(n_a k^2 + k^3)$; we develop it now.

4.2 Woodbury direct solve

The system matrix on the active set $\mathcal{A}$ is $C_{0,\mathcal{A}} = \bar{\mu} I_{n_a} + U_{k,\mathcal{A}} \Delta_k U_{k,\mathcal{A}}^\top$, where $U_{k,\mathcal{A}} = U_k[\mathcal{A}, :] \in \mathbb{R}^{n_a \times k}$ is the restriction of the signal eigenvectors to the active assets and $n_a = |\mathcal{A}|$. The Woodbury matrix identity [4], applied with $A = \bar{\mu} I_{n_a}$, $U = U_{k,\mathcal{A}}$, $C = \Delta_k$, $V = U_{k,\mathcal{A}}^\top$, gives

$$\big(C_{0,\mathcal{A}}\big)^{-1} b \;=\; \frac{b}{\bar{\mu}} \;-\; \frac{1}{\bar{\mu}^2}\, U_{k,\mathcal{A}}\, W^{-1} U_{k,\mathcal{A}}^\top b\,, \qquad W = \Delta_k^{-1} + \frac{1}{\bar{\mu}}\, U_{k,\mathcal{A}}^\top U_{k,\mathcal{A}} \;\in \mathbb{R}^{k \times k}\,.$$

Note that after restricting to $\mathcal{A}$ the columns of $U_{k,\mathcal{A}}$ are generally no longer orthonormal: $U_{k,\mathcal{A}}^\top U_{k,\mathcal{A}} \neq I_k$. The correction matrix W accounts for this; its Cholesky factorisation is used to evaluate $W^{-1} U_{k,\mathcal{A}}^\top b$. The active-set reduction requires $p+1$ such solves per outer step (the p columns of $B_{\mathcal{A}}^\top$ and $\hat{\mu}_{\mathcal{A}}$); all share the single factorisation of W, so each right-hand side beyond the first costs only $O(n_a k)$.

Proposition 4.1 (Woodbury solve complexity)**.** *One application of $(C_{0,\mathcal{A}})^{-1}$ costs:*

- *$O(n_a k)$ for the two matrix-vector products $U_{k,\mathcal{A}}^\top b$ and $U_{k,\mathcal{A}} z$;*
- *$O(n_a k^2)$ to form $W = \Delta_k^{-1} + \bar{\mu}^{-1} U_{k,\mathcal{A}}^\top U_{k,\mathcal{A}}$ (a rank-k outer product sum over n_a columns);*
- *$O(k^3)$ for the Cholesky factorisation of W.*

The dominant term is $O(n_a k^2)$; the total per-application cost is $O(n_a k^2 + k^3)$.

4.3 Algorithm and warm-starting

The tolerance $\varepsilon = 10^{-6}$ governs both the primal feasibility check (drop assets with $w_{\mathcal{A},i} < -\varepsilon$) and the dual violation threshold (admit assets with reduced gradient $r_i < -\varepsilon$); see Remark 4.2 for its scale. This is a standard KKT tolerance for active-set methods [17].

Two further features of Algorithm 1 deserve emphasis.

Algorithm 1 Woodbury portfolio solver

Require: Signal eigenpairs $(U_k, \Lambda_k) \in \mathbb{R}^{n\times k} \times \mathbb{R}^k$; bulk mean $\bar{\mu}$; balance system $B \in \mathbb{R}^{p\times n}$, $c \in \mathbb{R}^p$ *(default* $\mathbf{1}^\top$, 1*)*; return-tilt $\rho \geq 0$; expected returns $\hat{\mu}$; tolerance $\varepsilon = 10^{-6}$; optional warm start $(\mathcal{A}_0, w_0)$

Ensure: Portfolio weights $w \in \mathbb{R}^n$ with $w \geq 0$, $Bw = c$

$\mathcal{A} \leftarrow \mathcal{A}_0$ (or $\{1,\dots,n\}$ for cold start); $w \leftarrow w_0$ (or $\mathbf{0}$); $\Delta_k \leftarrow \Lambda_k - \bar{\mu}\mathbf{1}$
loop
  $U_a \leftarrow U_k[\mathcal{A}, :]$ ▷ $n_a \times k$ restriction
  $W \leftarrow \mathrm{diag}(1/\Delta_k) + U_a^\top U_a/\bar{\mu}$ *(*$O(n_a k^2)$*)*
  $L \leftarrow \mathrm{chol}(W)$; define $\mathrm{inv}(b) \leftarrow b/\bar{\mu} - U_a\,(L^\top\backslash(L\backslash U_a^\top b))/\bar{\mu}^2$ *(*$O(k^3)$ *once)*
  $V_1 \leftarrow \mathrm{inv}(B_{\mathcal{A}}^\top)$; $v_2 \leftarrow \mathrm{inv}(\hat{\mu}_{\mathcal{A}})$ if $\rho > 0$ *(*$p{+}1$ *solves,* $O(n_a k)$ *each)*
  $S \leftarrow B_{\mathcal{A}} V_1$; solve $S\lambda = 2c - \rho\, B_{\mathcal{A}} v_2$ *(*$p \times p$ *Schur,* $O(p^3)$*)*
  $w_{\mathcal{A}} = \frac{1}{2} V_1 \lambda + \frac{\rho}{2} v_2$
  if $\exists\, i \in \mathcal{A}$ with $w_{\mathcal{A},i} < -\varepsilon$ **then**
    $\mathcal{A} \leftarrow \mathcal{A} \setminus \{\arg\min_i w_{\mathcal{A},i}\}$ *(primal step; cf. Remark 4.1)*
  else
    $w \leftarrow 0$; $w[\mathcal{A}] \leftarrow w_{\mathcal{A}}$
    $g \leftarrow 2\big(\bar{\mu} w + U_k(\Delta_k(U_k^\top w))\big) - \rho\hat{\mu}$; $r \leftarrow g - B^\top\lambda$ *(reduced gradient,* $O(nk)$*)*
    $j^\star \leftarrow \arg\min_{i\notin\mathcal{A}} r_i$
    if $r_{j^\star} \geq -\varepsilon$ **then break** *(KKT satisfied)*
    end if
    $\mathcal{A} \leftarrow \mathcal{A} \cup \{j^\star\}$ *(dual step)*
  end if
end loop
return w

The return matrix never enters the solve. The dual-feasibility check evaluates the gradient of the objective, $g = 2\,C_0 w - \rho\hat{\mu}$, through the low-rank form at cost $O(nk)$. Neither $\hat{\Sigma}$ nor X is touched after preprocessing: the entire solve operates on the $O(nk)$ factors $(U_k, \Delta_k, \bar{\mu})$.

Warm-starting profits from active-set continuity, not initial guesses. A direct solver has no initial guess; warm-starting instead passes the previous active set and weights, cutting the number of outer steps. Along the frontier the Markowitz solution is piecewise-linear in ρ with generically single-asset breakpoint changes [15], so adjacent solves on a fine grid share all but a few active assets and the warm solver typically terminates in one to two outer steps; this is a structural property of parametric quadratic programming, not an artefact of the data.

Remark 4.3 (Relation to the critical line algorithm)**.** *The same piecewise-linear structure underlies the critical line algorithm (CLA) [15, 16, 20]. On a fixed active set the restricted solution and multipliers are affine in* ρ *(*$w_{\mathcal{A}}(\rho) = a + \rho b$*, with* $a = V_1 S^{-1} c$ *and* b *read off the same Schur solve), so every primal exit and dual entry is the root of an affine function and the exact frontier costs one Woodbury solve per breakpoint (Section 6.4). CLA enumerates the exact frontier for a* fixed *covariance, whereas the warm-started grid sweep extends unchanged to the practitioner's setting where consecutive solves differ in the data, not only in* ρ.

Remark 4.4 (Ridge terms and other structured modifications)**.** *The Woodbury structure survives several practically relevant modifications. A ridge penalty* $\gamma\|w\|^2$ *simply shifts the identity coefficient,* $\bar{\mu} \mapsto \bar{\mu} + \gamma$*. A diagonal loading* $D + U_k \Delta_k U_k^\top$ *with non-scalar* $D \succ 0$ *replaces* $b/\bar{\mu}$ *by* $D^{-1}b$ *at unchanged cost. Further* linear equality *constraints beyond the balance system (e.g. sector-neutrality)* *are absorbed by appending rows to B; the $p \times p$ Schur complement grows with them and the solve cost stays linear in p. What is* not *supported is a full-rank perturbation of the bulk spectrum, as produced by nonlinear shrinkage (Remark 2.2).*

4.4 Complexity and crossover analysis

Proposition 4.2 (Woodbury vs. KKT-Cholesky crossover)**.** *KKT-Cholesky assembles $C_{0,\mathcal{A}}$ by a rank-k update and factors it ($n_a^2 k + n_a^3/3$ flops); Woodbury forms and factors the $k \times k$ correction W ($n_a k^2 + k^3/3$ flops). Writing $x = k/n_a$, the Woodbury solve is cheaper per outer step iff $n_a k^2 + \frac{k^3}{3} < n_a^2 k + \frac{n_a^3}{3}$, which (dividing by n_a^3) factors as $(x-1)(x^2+4x+1) < 0$, i.e. iff $x < 1$ ($k < n_a$).*

Remark 4.5 (Crossover in practice)**.** *The crossover condition $k < n_a$ holds across essentially all of the frontier: on S&P 500 data $k = 17$ against $n_a = 46$ at the minimum-variance solution ($x \approx 0.37$), and on the synthetic frontier $x = k/n_a$ runs from about 0.33 at the high-return end ($n_a = 64$) to about 1 at the tightest (minimum-variance) point, where n_a dips to $\approx k = 21$. Only at that extreme are the two solves comparable; elsewhere the flop-count advantage of Proposition 4.2 applies, and the measured per-step gap is larger still ($5.7\times$ warm, Table 1), consistent with the Woodbury working set ($k \times k$) fitting in lower cache levels than the assembled $n_a \times n_a$ system, though we have not instrumented the memory hierarchy directly. When $n_a \le k$ the solver falls back to the KKT-Cholesky branch; the Woodbury solve itself remains valid for all $n_a \ge 1$, since $W = \Delta_k^{-1} + \bar{\mu}^{-1} U_{k,\mathcal{A}}^\top U_{k,\mathcal{A}}$ is the sum of a positive definite diagonal matrix and a positive semidefinite matrix, hence always invertible.*

Proposition 4.3 (Per-solve pipeline complexity)**.** *With s outer active-set steps, one solve costs $s \cdot O(n_a k^2 + k^3 + nk)$ (the $O(nk)$ term is the dual-step gradient). For a sweep of S solves sharing the eigenpairs, the pipeline cost is $O(nTk) + S\, s\, O(n_a k^2 + k^3 + nk)$, so the $O(nTk)$ randomised-SVD preprocessing (Proposition 5.1) amortises to $O(nTk/S)$ per solve.*

At the $n = 500$ benchmark the preprocessing share of a 50-point sweep is 30–45%; for a single solve it dominates. In every case the pipeline total stays well below the KKT-Cholesky solve alone at large n (Section 6.2).

5 Randomised SVD Preprocessing

The Woodbury solver requires the top-k eigenpairs of the sample correlation $C = D^{-1/2}\hat{\Sigma}\, D^{-1/2}$, equivalently of the standardised returns $X_s = XD^{-1/2}$. Two paths are available.

Dense path Assemble C explicitly ($O(n^2 T)$ FLOPs, $O(n^2)$ storage), then compute a full eigendecomposition ($O(n^3)$). Total cost $O(n^2 T + n^3)$; 0.011 s at $n = 494$, $T = 1213$ on the benchmark hardware.

Randomised SVD path The right singular vectors of $X_s/\sqrt{T}$ are the eigenvectors of C. A randomised truncated SVD [5] computes the top-k eigenpairs without forming $X_s^\top X_s$. The trace-preserving bulk mean is recovered without any dense decomposition from $\bar{\mu} = \big(n - \sum_{j \le k} \hat{\lambda}_j\big)/(n-k)$, using $\mathrm{tr}(C) = n$.

Proposition 5.1 (Randomised SVD cost and accuracy)**.** *Algorithm 2 costs $O(nT(k+p))$ FLOPs and $O(n(k+p))$ storage; no $n \times n$ matrix is formed. Let $P = QQ^\top$ denote the orthogonal projector onto the computed basis. Since the sketch $Y = T\,\hat{\Sigma}\,\Omega$ applies the range finder of [5] to the symmetric*

Algorithm 2 Randomised truncated SVD for eigenpairs of the correlation C

Require: Standardised returns $X_s = XD^{-1/2} \in \mathbb{R}^{T\times n}$; target rank k; oversampling $p \geq 5$
Ensure: Signal eigenvectors $U_k \in \mathbb{R}^{n\times k}$, eigenvalues $\Lambda_k \in \mathbb{R}^k$, bulk mean $\bar{\mu}$
1: $\Omega \leftarrow \mathcal{N}(0, I_{n\times(k+p)})$ ▷ random test matrix
2: $Y \leftarrow X_s^\top (X_s\Omega)$ ▷ $n \times (k+p)$; two matrix-vector products with X_s, $O(nT(k+p))$
3: $Q \leftarrow \mathrm{orth}(Y)$ ▷ orthonormal basis, $O(n(k+p)^2)$
4: $B \leftarrow (X_sQ)^\top (X_sQ)/T$ ▷ $(k+p)\times(k+p)$ projected correlation; no $X_s^\top X_s$ formed
5: $[\hat{U}, \hat{\Lambda}, \sim] \leftarrow \mathrm{eigh}(B)$; sort descending
6: $U_k \leftarrow Q\hat{U}[:, 1{:}k]$; $\Lambda_k \leftarrow \hat{\Lambda}[1{:}k]$
7: $\bar{\mu} \leftarrow \big(n - \sum_{j\leq k} \hat{\Lambda}_j\big)/(n-k)$; identify k via MP threshold
8: **return** $(U_k, \Lambda_k, \bar{\mu})$

matrix $\hat{\Sigma}$ with Gaussian test matrix Ω and oversampling $p \geq 2$, Theorem 10.6 of [5] gives, in expectation,

$$\mathbb{E}\,\big\|(I-P)\hat{\Sigma}\big\|_2 \;\leq\; \Big(1 + \sqrt{\tfrac{k}{p-1}}\Big)\,\lambda_{k+1} \;+\; \frac{e\sqrt{k+p}}{p}\Big(\textstyle\sum_{j>k} \lambda_j^2\Big)^{1/2} \;=:\; \epsilon_{k,p}\,,$$

where λ_j are the eigenvalues of $\hat{\Sigma}$. The returned eigenpairs are the exact eigenpairs of the compression $P\hat{\Sigma}P$, and by Weyl's inequality the eigenvalue errors satisfy $|\lambda_j - \hat{\lambda}_j| \leq \|\hat{\Sigma} - P\hat{\Sigma}P\|_2 \leq 2\,\|(I-P)\hat{\Sigma}\|_2$ for all $j \leq k$.

Proof. The cost and storage counts follow from the line-by-line annotations of Algorithm 2. The error bound is Theorem 10.6 of [5] applied to $A = \hat{\Sigma}$ (the deterministic scaling T of the sketch does not change its range). For the compression bound, write $\hat{\Sigma} - P\hat{\Sigma}P = (I-P)\hat{\Sigma} + P\hat{\Sigma}(I-P)$ and note that both terms have spectral norm at most $\|(I-P)\hat{\Sigma}\|_2$ by symmetry; Weyl's inequality [6] then bounds each eigenvalue perturbation by the norm of the difference. □

The bound $\epsilon_{k,p}$ is governed by the correlation eigenvalues at and below the MP edge $\lambda_{k+1} \approx \hat{\lambda}_+ \approx 2.28$ on S&P 500 data, against a leading signal eigenvalue $\lambda_1 \approx 313\,\bar{\mu}$ that is two orders of magnitude larger. The bound is conservative (the tail-sum term accumulates all $n-k$ bulk eigenvalues) and the measured accuracy is far better.

Memory passes Algorithm 2 involves four matrix-vector products with X in total: two for the sketch $Y = X^\top(X\Omega)$ (line 2) and two for the projection $B = Q^\top X^\top XQ$ (line 4, computed as $(XQ)^\top(XQ)/T$). Reading X from memory is required at two distinct stages: once for the sketch (line 2) and once for the projection (line 4). For very large n and T where X does not fit in memory, a single-pass variant that combines both stages is available at a modest increase in approximation error; see [5] for details. At benchmark size ($n = 494$, $T = 1213$, approximately 4.6 MB for double-precision X) both stages fit comfortably in L3 cache.

Benchmark On S&P 500 ($n = 494$, $T = 1213$, $k = 17$, $p = 10$) the randomised SVD is 1.9× faster than the dense path (0.006 s vs 0.011 s), needs 29× less storage, and aligns the top-k signal subspace to within a Frobenius error $\|UU^\top - \hat{U}\hat{U}^\top\|_F \approx 2.5 \times 10^{-2}$ (about 0.6% per eigenvector column), an order of magnitude below the $k \pm 1$ threshold uncertainty (Section 6.5) that any user of the MP rule already accepts. The advantage grows with n: at $n = 3000$ the dense eigendecomposition takes 1.12 s against 0.056 s for rSVD, a 20× saving (Figure 1).

Large-n recommendation For $n \leq 500$ either path is practical; the dense path is simpler to implement and both finish in milliseconds. For $n > 1000$ the randomised SVD is strongly preferred: the dense path's $O(n^2)$ storage requirement alone (8 MB at $n = 1000$, 72 MB at $n = 3000$) begins to stress L3 cache, while the randomised path's $O(n(k+p))$ requirement (about 3 MB at $n = 3000$, $k = 125$) stays within typical L3.

6 Numerical Results

All experiments were run on an Apple M4 Pro (14-core CPU, 48 GB RAM) under Python 3.12 with NumPy 2.4 (Apple Accelerate BLAS), SciPy 1.17, and scikit-learn 1.x. Reported times are the minimum of three repeated runs, the standard protocol for suppressing scheduling noise; the spread across repetitions was below ten percent for all entries. The solver comparison and scaling experiments (Sections 6.1 and 6.2) use synthetic returns drawn from a spiked covariance model calibrated to the S&P 500 regime ($n/T \approx 0.4$, factor structure matching the Marchenko–Pastur prediction; seed fixed at 42). The condition-number and k-sensitivity results (Sections 2 and 6.5) use the actual S&P 500 return series ($n = 494$, $T = 1213$); the out-of-sample backtests (Section 6.6) additionally use the FTSE 100 series.

6.1 Solver comparison

Two solvers are applied to the same problem: minimise $w^\top \Sigma_0 \, w - \rho \, \hat{\mu}^\top w$ subject to $w \geq 0$, $\mathbf{1}^\top w = 1$ (the budget case) over a 50-point efficient frontier. Synthetic equity data: $n = 500$ assets, $T = 1250$ days, $k = 22$ signal factors. Active-set sizes averaged $n_a = 42$ across the frontier (warm start).

- **KKT-Cholesky**: the same primal-dual outer loop as Algorithm 1, but with the inner solve replaced by explicitly assembling $T_{0,\mathcal{A}}^{\mathrm{RMT}}$ as a dense $n_a \times n_a$ matrix via a rank-k symmetric update and then factoring by Cholesky.
- **Woodbury**: Algorithm 1 as stated: the inner solve uses the Woodbury identity and never assembles an $n_a \times n_a$ matrix.

	Cold start		Warm start	
Solver	Total (s)	ms/pt	Total (s)	ms/pt
Synthetic, $n = 500$, $T = 1250$, $k = 21$, 50-point sweep (total / ms per point)				
KKT-Cholesky ($k = 21$)	0.8830	17.7	0.0668	1.3
Woodbury ($k = 21$)	0.0586	1.2	0.0117	0.2
S&P 500, $n = 494$, $T = 1213$, $k = 17$, single min-var solve (seconds / ms)				
KKT-Cholesky ($k = 17$)	0.0183	18.3	0.0016	1.6
Woodbury ($k = 17$)	0.0011	1.1	0.0001	0.1

Table 1: Solver timings on two datasets. *Top panel*: synthetic equity data, 50-point frontier sweep; totals and ms per frontier point. *Bottom panel*: real S&P 500 data ($n = 494$, $T = 1213$), single minimum-variance solve; the 'ms/pt' column gives ms per solve. Woodbury and KKT-Cholesky produce identical portfolios to machine precision.

Correctness The active-set loop terminates only on a verified KKT certificate (the $N = 0$ exit of Algorithm 1), so each returned portfolio is the global optimum of its equality-constrained subproblem up to the tolerance ε. Woodbury and KKT-Cholesky execute the same outer loop on the same subproblems and differ only in the inner linear algebra, so they return identical portfolios: the maximum frontier-volatility difference across the sweep is 9×10^{-15} percentage points, confirming that the Woodbury inverse reproduces the dense Cholesky solve to machine precision.

Timings On synthetic data (top panel), Woodbury is 15× faster than KKT-Cholesky on a cold start and 5.7× faster warm. Both methods execute the same outer active-set loop; the difference is entirely in the inner solve. KKT-Cholesky assembles the $n_a \times n_a$ active-set matrix at each outer step via a rank-k symmetric update and factors by Cholesky (mean $n_a = 31$ over the frontier). Woodbury instead operates on the $k \times k = 21 \times 21$ correction matrix W (Remark 4.5). The cold-start ratio exceeds the warm-start ratio because the cold solver takes more outer active-set steps before converging at systematically larger active-set sizes, where the per-step gap between $O(n_a^2 k + n_a^3)$ and $O(n_a k^2)$ is widest.

The bottom panel confirms the result on real S&P 500 data ($n = 494$, $T = 1213$, $k = 17$, $n_a = 46$ active assets at the minimum-variance solution): Woodbury is about 17× faster than KKT-Cholesky on a single cold minimum-variance solve. For the single-solve setting the warm start uses the converged active set from an immediately preceding cold solve at the same $\rho = 0$, measuring the incremental re-solve cost when the covariance is refreshed but the optimal active set has not changed, the scenario for daily rebalancing. The warm-start advantage is not an artefact of the synthetic data-generating process: the Markowitz frontier is piecewise linear in weight space with generically single-asset exchanges at breakpoints (Section 4), for any return distribution.

6.2 Scaling with n

Figure 1 shows wall-clock time vs. n ($T = 1250$ fixed) for four configurations: KKT-Cholesky (no preprocessing), Woodbury solve only, Woodbury with dense preprocessing, and Woodbury with randomised SVD preprocessing.

Two caveats frame this experiment. First, the growth law $k \approx n/24$ is observed on the real data at a single size ($k = 17$ at $n = 494$) and is *built into* the synthetic generator; the large-n conclusions are conditional on it. Second, under that law both methods scale cubically (Cholesky as $O(n_a^3)$ with $n_a \propto n$, Woodbury as $O(n_a k^2)$ with $k \propto n$), so the asymptotic gap is a constant factor, not an order. The *measured* solve-only ratio nevertheless grows from 15× at $n = 500$ to 53× at $n = 3000$, consistent with the $n_a \times n_a$ working set falling out of cache while the $k \times k$ correction matrix does not. If k is fixed externally as n grows (Remark 2.3), Woodbury gains a full asymptotic order over Cholesky.

Randomised SVD preprocessing is competitive throughout. At $n = 3000$ the dense eigendecomposition takes 1.12 s, against 0.056 s for the rSVD path (a 20× saving) and 0.028 s for the Woodbury solve; the full pipeline (rSVD + Woodbury, 0.084 s) is 18× faster than the KKT-Cholesky solve alone (1.52 s). Dense preprocessing is impractical for $n > 750$. For $S \geq 2$ solves (a frontier sweep, a rebalance sequence) the preprocessing cost amortises further, so the pipeline advantage grows with the number of solves.

6.3 Efficient frontier

Figure 2 shows the 50-point efficient frontier on synthetic equity data ($n = 500$, $T = 1250$), coloured by active-asset count. The minimum-variance point (star) has $n_a = 20$ active assets; the high-return

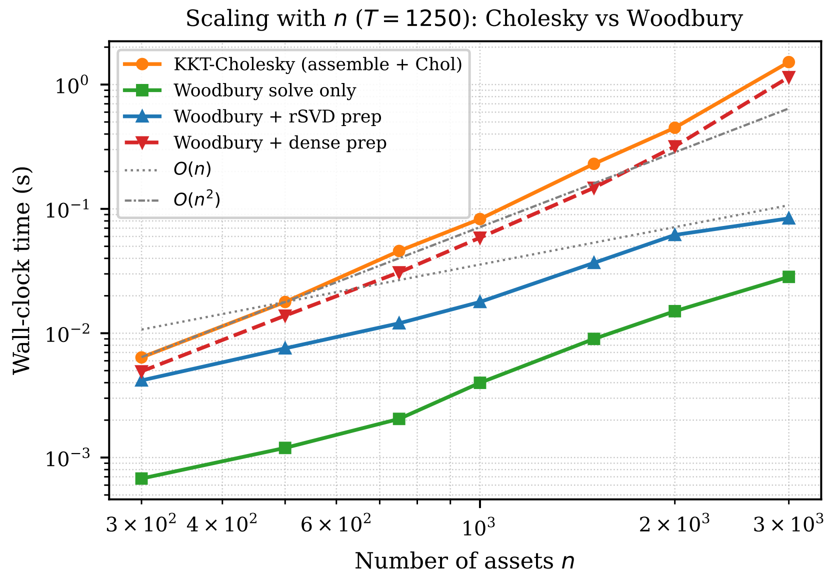

Figure 1: Wall-clock time vs. n (log-log, $T = 1250$; k grows as $\approx n/24$). KKT-Cholesky grows faster than Woodbury as n_a grows with n and the $O(n_a^3)$ term dominates; dashed reference lines $O(n)$, $O(n^2)$ anchored at $n = 500$. At $n = 3000$ ($k = 125$) the Woodbury solve is 53× faster than KKT-Cholesky, and the full rSVD-plus-Woodbury pipeline (0.084 s) is 18× faster than KKT-Cholesky (1.52 s), which is charged no preprocessing.

end has $n_a = 64$. The rapid decrease in active assets away from the mean-variance efficient region is characteristic of the long-only constraint and explains the warm-start efficiency: adjacent frontier points share all but a few active assets.

6.4 Exact frontier via the critical line algorithm

On each active set the restricted solution is affine in ρ (Remark 4.3), so the exact efficient frontier can be traced breakpoint-to-breakpoint with a single Woodbury solve per segment, at a cost comparable to the coarse grid sweep but returning the complete piecewise-affine path rather than a fixed number of samples. For a fixed covariance the exact sweep is preferable; the warm-started grid sweep remains the tool of choice when consecutive solves differ in the data rather than only in ρ.

6.5 Threshold Robustness and k-Sensitivity

The Marchenko–Pastur threshold determines k from data. A practical concern is whether the portfolio is sensitive to this choice: a threshold that incorrectly includes one extra noise factor (or excludes one signal factor) by a small amount would be problematic if the resulting portfolio changes substantially.

6.5.1 Cost of auditing

Auditing the sensitivity at $k \pm 1$ requires two additional Woodbury solves. Since the eigenpairs (U_k, Λ_k) are already computed, these are pure solve costs at $O(n_a k^2 + k^3)$ each, negligible relative to the preprocessing. On the benchmark dataset, the three-point audit ($k = 16, 17, 18$) adds less than 3 ms to the total pipeline. This makes the audit a costless safety check.

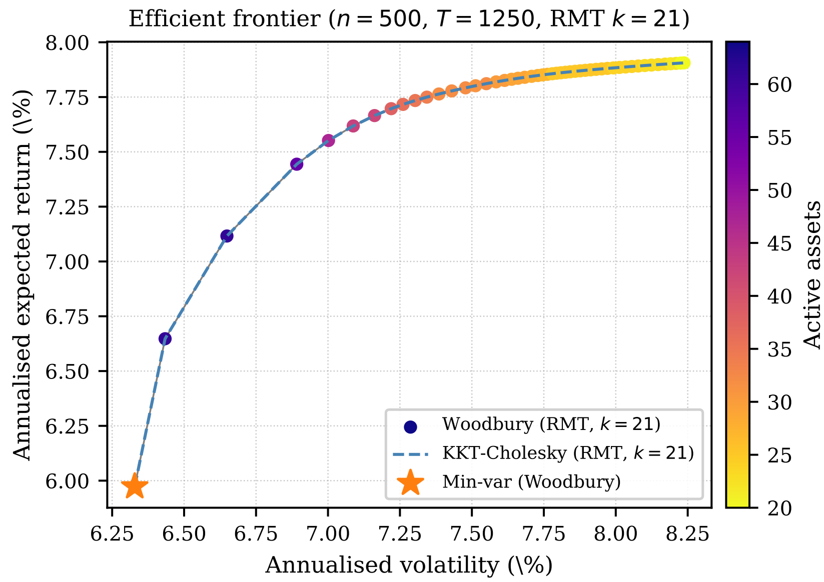

Figure 2: Efficient frontier ($n = 500$, $T = 1250$, $k = 21$, $\alpha = 1$). Scatter: Woodbury warm, coloured by active-asset count (star = minimum-variance portfolio, $n_a = 20$). Dashed: KKT-Cholesky warm (same portfolio to numerical precision, plotted to confirm equivalence). Both sweeps take under 0.02 s warm; see Table 1 for timings.

6.5.2 Analytical sensitivity

Adding the $(k+1)$-th factor is a rank-one update $C_0(k+1) \approx C_0(k) + \delta_{k+1} u_{k+1} u_{k+1}^\top$ with $\delta_{k+1} = \lambda_{k+1} - \bar{\mu} > 0$, so by Sherman–Morrison the portfolio perturbation is controlled by δ_{k+1}, the eigenvalue excess of the marginal factor. Since λ_{k+1} sits just above the edge $\hat{\lambda}_+$, δ_{k+1} is small relative to the signal excesses and the perturbation is correspondingly small; the empirical audit below includes the exact bulk-mean adjustment.

6.5.3 Empirical sensitivity on S&P 500 data

Table 2 quantifies the change for $k \in \{16, 17, 18\}$ on the full S&P 500 sample ($n = 494$, $T = 1213$). The maximum absolute weight change across all assets is 33 bp when one signal factor is removed ($k = 16$) and 104 bp when one is added ($k = 18$); the ℓ_2 weight-vector distances are 77 bp and 233 bp respectively. The annualised volatility of the minimum-variance portfolio changes by at most a few basis points in either direction.

k	Active	$\mathrm{Vol}_{\mathrm{ann}}$ (%)	$\|w(k) - w(k^*)\|_\infty$	$\|w(k) - w(k^*)\|_2$
16	47	10.40	0.0033	0.0077
17	46	10.40	0.0000	0.0000
18	45	10.38	0.0104	0.0233

Table 2: Portfolio sensitivity to k at $\rho = 0$ on S&P 500 data ($n = 494$, $T = 1213$). $k^* = 17$ is the MP threshold. All three Woodbury solves complete in under 3 ms combined (preprocessing amortised). Maximum absolute weight change: 104 bp.

These results indicate that the MP threshold is not a sensitive hyperparameter for the risk objective: misjudging k by one factor moves individual weights by up to about a percentage point but changes portfolio volatility by only a few basis points. A practitioner unsure whether the correct k is 16, 17, or 18 can audit all three at negligible cost.

6.6 Out-of-sample performance

The speed results above are only useful if the RMT-cleaned estimator is statistically sound on real data. We therefore run rolling-window long-only minimum-variance backtests on two universes: the S&P 500 ($n/L \approx 0.98$, deliberately high-dimensional) and the FTSE 100 ($n/L \approx 0.17$, a comfortably sampled regime in which cleaning should matter less). In both cases: estimation window $L = 504$ days, monthly rebalancing, buy-and-hold drift between rebalances. Assets with zero full-sample variance (delisted names forward-filled flat, late listings padded with zeros) are excluded: a zero-variance asset would absorb the entire minimum-variance portfolio. Four strategies are compared: the RMT-CRE portfolio of this paper ($\alpha = 1$, cleaned on the correlation and solved by Algorithm 1); linear shrinkage to the scaled identity at the practitioner level $\alpha = 0.5$ and at the Ledoit–Wolf oracle intensity [9]; and equal weight.

Strategy	OOS vol (% ann.)	Sharpe	Turnover (%)
S&P 500 ($n = 494$, median $k = 12$)			
RMT ($\alpha = 1$)	9.96	0.261	12.9
LW-0.5 ($\alpha = 0.5$)	9.74	-0.022	9.1
LW-oracle	9.75	0.033	13.9
Equal-weight	14.54	0.256	3.0
FTSE 100 ($n = 86$, median $k = 5$)			
RMT ($\alpha = 1$)	11.28	-0.649	9.4
LW-0.5 ($\alpha = 0.5$)	10.67	-0.456	5.5
LW-oracle	10.79	-0.609	9.1
Equal-weight	14.30	-0.119	2.6

Table 3: Out-of-sample rolling-window backtests (estimation window $L = 504$ days, monthly rebalancing). Turnover is the mean one-way turnover per rebalance. The per-window MP-detected k is shown in each panel header.

The pattern is consistent across both universes. All three minimum-variance strategies cut realised volatility by a fifth to a third relative to equal weight. On the S&P 500 the RMT-CRE portfolio (9.96%) sits within about 0.2 percentage points of the linear-shrinkage alternatives (9.75% and 9.74%) with comparable turnover; on the FTSE 100 it is within roughly half a point (11.28% against 10.79% and 10.67%). CRE does not beat linear shrinkage on either universe, but the gap is within the noise of a single backtest, and CRE uses no tuning parameter beyond the MP threshold. We claim only that the estimator whose structure enables the direct solver carries no material out-of-sample penalty, consistent with the comparative literature on RMT cleaning [2].

Reproducibility All experiments are produced by a single script (`experiment_rmt.py`) in the repository of the accompanying open-source package; random seeds are fixed (42 for the synthetic generator, 0 for the randomised SVD), and the S&P 500 and FTSE 100 return panels are included in the repository.

7 Conclusions

The sample covariance of equity returns mixes a small number of well-estimated signal directions with several hundred poorly estimated noise directions in its correlation structure. Cleaning the sample correlation by the trace-preserving CRE rule of Marchenko–Pastur analysis [8, 2] cuts the correlation condition number by a factor of roughly 290 on S&P 500 data (to $\kappa \approx 313$); its scalar-identity-plus-low-rank form admits an exact direct solve via the Woodbury identity once the problem is posed in the standardised coordinates $y = D^{1/2}w$.

This paper makes three algorithmic contributions and one empirical contribution.

Direct Woodbury portfolio solver. In the standardised coordinates the primal-dual active-set loop requires an inner solve of $C_{0,\mathcal{A}}\, y = b$ at each outer step, the p balance constraints eliminated by a $p \times p$ Schur complement. The naive approach assembles the $n_a \times n_a$ active-set matrix and solves by Cholesky at $O(n_a^2 k + n_a^3)$ per step; the Woodbury identity reduces this to $O(n_a k^2 + k^3)$ by working with the $k \times k$ correction matrix, cheaper whenever $k < n_a$ (Proposition 4.2), which holds across essentially all of the long-only frontier. The measured wall-clock advantage is 5.7× warm and 15× cold against KKT-Cholesky on a 50-point synthetic sweep, and about 17× cold on a real S&P 500 minimum-variance solve; the returned weights satisfy the KKT conditions and match the KKT-Cholesky solution to machine precision. The same kernel supplies an exact critical-line sweep (Section 6.4).

Randomised SVD preprocessing. The top-k correlation eigenpairs are computed by a randomised truncated SVD applied directly to the standardised returns in $O(nTk)$ time and $O(nk)$ storage, without forming $\hat{\Sigma}$. At $n = 3000$ this is 20× faster than a dense eigendecomposition, and the full pipeline (rSVD + Woodbury) is 18× faster than the KKT-Cholesky solve alone; for a sweep of S solves the preprocessing amortises to $O(nTk/S)$.

Scope of the structural trick. Ridge terms, diagonal loadings, externally specified factor models, and linear equality constraints preserve the Woodbury structure (Remark 4.4); nonlinear shrinkage and general rotationally invariant estimators [11, 12, 2] do not, because they modify the full spectrum. CRE clipping is the member of the RMT cleaning family that trades a known statistical refinement for an exploitable algebraic structure.

Empirical validation. The MP threshold is an insensitive hyperparameter: misjudging k by one factor moves portfolio weights by tens of basis points and volatility by under a basis point, and the three-point audit costs milliseconds. Rolling out-of-sample backtests on the high-dimensional S&P 500 ($n/L \approx 0.98$) and the well-sampled FTSE 100 ($n/L \approx 0.17$) place the CRE portfolio within about half a volatility point of linear shrinkage on both universes, with all minimum-variance strategies materially less volatile than equal weight: the estimator that enables the fast solver carries no material out-of-sample penalty.

Practical recommendations For $n \leq 500$ either preprocessing path is practical; use dense if simplicity matters. For $n > 1000$, use randomised SVD. For any sweep of $S \geq 2$ solves (efficient frontier, rebalance sequence, sensitivity audit) the preprocessing cost amortises to $O(nTk/S)$ per solve. The Woodbury solver is preferred over KKT-Cholesky whenever $k < n_a$, which holds across essentially all of the long-only institutional equity regime.

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